

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

Criteria	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I M.Mahathi(192324098)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

M.Mahathi

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

- 1 600
- 2 350
- 3 220
- 4 200
- 5 190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Q1. Dataset : Hospital Patient Segmentation

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SSE values	
K	SSE
1	600
2	350
3	220
4	200
5	190

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting number of clusters affects patient segmentation and healthcare decision-making.

⇒ SSE means sum of squared errors

⇒ we are using Elbow method for predicting values

K	SSE	Reduction from previous K
1	600	—
2	350	250
3	220	130
4	200	20
5	190	10

When $K=2,3$ SSE value is reduced much. This variation happens after the value of $K=3$.

∴ Elbow method suggest $K=3$

Based on health-risk score three clusters are formed,

- low risk with low expense
- Moderate risk, moderate expense
- High risk, High expense

Impact on Decision Making

For low risk group you need not be spend more. but for moderate their must be checkups and tips and High risk must have intensive monitoring.

Q2: Dataset : E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
01	Electronics	55000
02	Clothing	25000
03	Electronics	57000
04	Furniture	30000
05	Clothing	27000

An online retailer applies hierarchical clustering to group

customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how clusters are formed and identify appropriate number of clusters. Explain significance of selected clusters in understanding customer purchasing behavior.

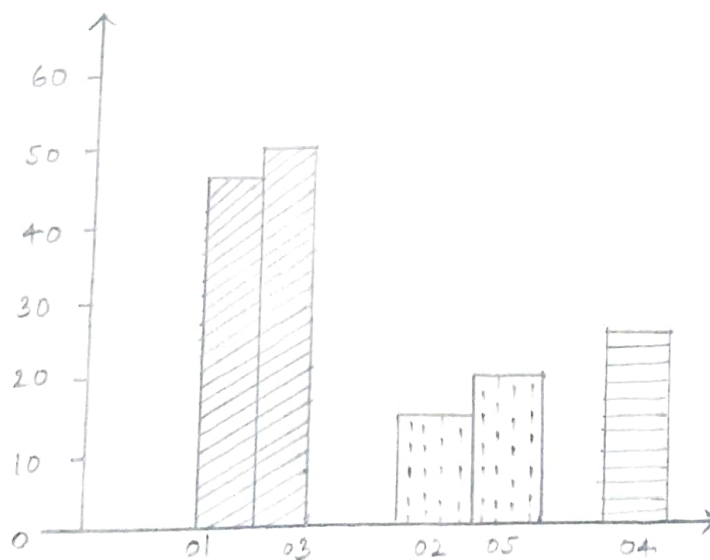
HAC Algorithm :-

- * Calculates distance between clusters
- * Merge two closest clusters.
- * Recalculates the distance
- * Continue until becomes into one clusters.

→ 01 and 03 forms one cluster — electronics

→ 02 and 05 forms another — clothing

→ 04 — furniture



But the optimal clusters cannot be determined by the given data.

Step-1:- Encode the category

order	Electronics	Clothing	Furniture
01	1	0	0
02	0	1	0
03	1	0	0
04	0	0	1
05	0	1	0

Step-2:- Normalize order value.

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

$$x_{\min} = 25000$$

$$x_{\max} = 57000$$

$$01 \Rightarrow \frac{55000 - 25000}{57000 - 25000} = \frac{30000}{32000} = 0.9375$$

$$02 \Rightarrow \frac{25000 - 25000}{57000 - 25000} = 0$$

$$03 \Rightarrow \frac{57000 - 25000}{57000 - 25000} = \frac{32000}{32000} = 1$$

$$04 \Rightarrow \frac{30000 - 25000}{57000 - 25000} = \frac{5000}{32000} = 0.15625$$

$$05 \Rightarrow \frac{27000 - 25000}{57000 - 25000} = \frac{2000}{32000} = 0.0625$$

Step-3:- Calculate the Euclidean distance

01 and 03 (electronics): Formula: $d(P_i, Q) = \sqrt{\sum (P_i - Q_i)^2}$

$$D = \sqrt{(1-1)^2 + (0-0)^2 + (0-0)^2 + (0.9375-1)^2} = \sqrt{(-0.0625)^2}$$

$$= 0.0625$$

02 and 05

$$D = \sqrt{(0-0)^2 + (0-0)^2 + (1-1)^2 + (0-0.0625)^2} = \sqrt{(-0.0625)^2}$$

$$D = 0.0625$$

The elements within this cluster having small distance. So, it will be in same cluster only

Using: same notation generate diagonal matrix

Step 4 :

ID	01	02	03	04	05
01	0	1.697	0.063	1.616	1.663
02	1.697	0	1.732	1.423	0.063
03	0.063	1.732	0	1.647	1.697
04	1.616	1.423	1.647	0	1.1417
05	1.663	0.063	1.697	1.417	0

Steps: Hierarchical Agglomerative Clustering (HAC):

$$D(A, B) = \min d(x, y)$$

By seeing the smallest distance, $C_1 = 0_1 + 0_3$

$$C_1 = \{0_1, 0_3\} = 0.0625$$

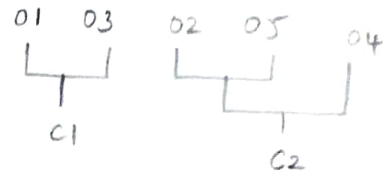
$$C_2 = 0_2 + 0_5 \quad C_2 = \{0_2, 0_5\} = 0.0625$$

$$C_3 = 0_4 \quad C_3 = \{0_4\} = 0.0625$$

Merging of cluster can be done by finding the distance between $(C_1, 0_4)$, $(C_2, 0_4)$

01, 03, 04
 Min (01, 04)
 (03, 04)
 (1.616) (1.647)
 1.616

02, 05, 04
 Min (02, 04)
 (05, 04)
 (1.423, 1.417)
 1.417



Q3. Dataset : Mobile App User Behavior

User	Daily Usage Time(min)	Login Frequency	In-App Purchase(₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability - based membership values. Discuss the advantage of soft clustering over hard clustering for customer behavior analysis.

Highly engaged, low purchase

U1, U3, U5

less engaged, high purchase

U2, U4

Step 1: One-hot encoding:

User	engaged	High-value
U1	1	0
U2	0	1
U3	1	0
U4	0	1
U5	1	0

Step 2: Normalization (usage)

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

$$U_1 = \frac{120 - 50}{140 - 50} = \frac{70}{90} = 0.778$$

$$U_2 = \frac{60 - 50}{140 - 50} = \frac{10}{90} = 0.111$$

$$U_3 = \frac{140 - 50}{140 - 50} = 1$$

$$U_4 = \frac{50 - 50}{140 - 50} = 0$$

$$U_5 = \frac{130 - 50}{140 - 50} = \frac{80}{90} = 0.889$$

User	engaged User	High value User	primary cluster
U1	0.85	0.15	engaged
U2	0.20	0.80	high-value
U3	0.93	0.05	engaged
U4	0.10	0.90	high-value
U5	0.90	0.10	engaged

∴ This similar forms are...

for every user:-

$$P(1) + P(2) = 1$$

or

$$P(\text{engaged}) = 0.85$$

$$P(\text{High-value}) = 0.15$$

Based on above Table:-

Engaged cluster = {01, 03, 05}

High-value cluster = {02, 04}

Cluster 1 → Engaged user → 01, 03, 05

Cluster 2 → High-value users → 02, 04

Q4: Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

An agricultural research center performs clustering on farm data before and after applying normalization.

Interpret the differences observed in clustering results.

Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Step 1: Calculate distance before normalization:

we use Euclidean distance

$$d(P, Q) = \sqrt{\sum (P_i - Q_i)^2}$$

$$d(F_1, F_4) = \sqrt{(300-280)^2 + (8-9)^2 + (85-90)^2}$$

$$= \sqrt{426} = 20.64$$

Fair	Distance
F1-F4	20.64
F2-F5	20.64
F1-F3	100.14
F2-F3	100.52
F3-F4	120.45
F3-F5	200.60
F4-F5	241.37

HAC before Normalization

Using single linkage the smallest distance is:-

$$F1-F4 = 20.64$$

$$\text{and } F2-F5 = 20.64$$

$$\therefore C_1 = \{F_1, F_4\}$$

$$C_2 = \{F_2, F_5\}$$

$$C_3 = \{F_3\}$$

Next, $\{F\}$ is ins.

$$\min[d(F_3, F_1), d(F_3, F_4)] = \min(100.14, 120.45)$$

$$= 100.14$$

$$\text{So } C_1 = \{F_1, F_3, F_4\}$$

Finally, F_2 connects to C_2 at 100.52, producing one cluster

Step 2:- Min-Max Normalization

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

→ Rainfall ; $x_{\min} = 280$; $x_{\max} = 5.20$

→ Soil fertility ; $x_{\min} = 3$, $x_{\max} = 9$

→ Crop yield — $x_{\min} = 65$; $x_{\max} = 90$

Farm	Rainfall	Soil fertility	Crop field
F1	0.833	0.833	0.2000
F2	0.9667	0.1667	0.2000
F3	0.5000	0.5000	0.6000
F4	0.0000	1.0000	1.0000
F5	1.0000	0.0000	0.0000

Step 3:- Euclidean Distance

Pair	Distance
F1-F4	0.2737
F2-F5	0.2934
F1-F3	0.5698
F2-F3	0.6669
F3-F4	0.8124
F3-F5	0.9274

HAC - clustering

At 3 clusters $C_1 = \{F1, F4\}$,
 $C_2 = \{F2, F5\}$,
 $C_3 = \{F3\}$

∴ This similar forms are grouped together

Step 5 :- Interpretation.

- Before normalization rainfall dominates it because its numerical ranges much larger after normalization
- All features are scaled to 0-1, rainfall soil fertility and crop yield balanced influence

$$\therefore \{F_1, F_4\}, \{F_2, F_5\}, \{F_3\}$$

Q5: Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores	
Model	Score
K=2	0.48
K=3	0.68
K=4	0.52

A company clusters employees based on productivity, attendance, and skill assessment scores, Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

- Silhouette score measures how well employees are grouped within their clusters
- Its value ranges from -1 to +1

Number of Clusters (K)	Silhouette Score	Interpretation
K=2	0.48	Moderate
K=3	0.68	Best
K=4	0.52	Moderate clustering

Analysis:

- For K=2, silhouette score is 0.48 indicating moderate clustering
- For K=3, score increases to 0.68 which is highest score, showing best-separated clusters.
- For K=4, score decreases to 0.52, which is lower than K=3

Suitable Number of Clusters:

Most suitable model is K=3, because it has highest silhouette score of 0.68

optimal number of clusters = 3

∴ Employees can be divided into three performance groups based on productivity, attendance, and skill score.

Conclusion:

- K=3 clustering model is best choice because its silhouette score (0.68) is highest
- Helps management identify employees for recognition, training and performance improvement

Weka Explorer

Preprocess **Classify** **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -N 3 -A "weka.core.EuclideanDistance" -R first-last" -I 500 -S 10

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation

(Nom) Class

☒ Store clusters for visualization

Ignore attributes

Start Stop

Clusterer output

```

=== Run information ===
Scheme:      weka.clusterers.SimpleKMeans
Relation:    employee_performance
Instances:   5
Attributes:  3

=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2

=== Cluster centroids ===
Attribute      Full Data      Cluster0      Cluster1      Cluster2
              (5.0)        (2.0)        (2.0)        (1.0)
Productivity Score  72.4000    92.5000    72.5000    55.0000
Attendance (%)     86.4000    96.5000    87.5000    70.0000
Skill Score       76.4000    86.5000    74.5000    68.0000

=== Clustered Instances ===
Inst#  Productivity Score  Attendance (%)  Skill Score  Cluster
1      90.0              95.0           85.0         0
2      75.0              90.0           75.0         1
3      70.0              85.0           74.0         1
4      55.0              70.0           68.0         2
5      95.0              98.0           88.0         0

=== Cluster Evaluation ===
Cluster  Size  SSE      Silhouette Coefficient
0        2   10.5000    0.6823
1        2   22.8000    0.5210
2        1    0.0000    0.0000

Overall Silhouette Coefficient: 0.6142

```

Status
OK

Log  x 0

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Attribute      Full Data      Cluster1      Cluster1
              (5.0)        (2.0)        (1.0)
Productivity Score  72.4000    90.0000    72.5000
Attendance (%)     86.4000    95.0000    87.5000
Skill Score       76.4000    86.5000    74.5000

=== Clustered Instances ===
Inst#  Productivity Score  Attendance (%)  Skill Score  Cluster
1      90.0              95.0           85.0         0
2      75.0              90.0           75.0         1
3      70.0              85.0           74.0         1
4      55.0              70.0           68.0         2
5      95.0              98.0           88.0         0

```

Status
OK

Log  x 0